X-Sense: An Explainable Multimodal and Real-Time Sentiment Intelligence System for Multilingual Social Media Streams

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Abstract

The widespread use of social media across different languages and cultures has led to a massive amount of user-created content. The emotional aspect of this content holds great value for businesses, researchers, policymakers and public health professionals. Even though there have been a major advancement in computational sentimental analysis the systems currently in use still have three major issues: they only use one type of input, they don't work well with languages other than English, and they can't explain why a particular result was given. This paper introduces Stage One of the X-Sense project, a system that aims to provide explainable, multimodal, and real-time sentiment classification for social media content in English, Hindi, and Marathi. After reviewing over forty studies from 2013 to 2024, five main types of computational approaches were examined and compared: methods based on lexicons, classical supervised classifiers, deep sequential and convolutional networks, transformer-based pre-trained models, and multimodal fusion architectures. The role of Explainable AI in creating user-interpretable explanations for predictions was also explored. The analysis showed that existing systems meet at most two out of the five key design requirements that define X-Sense: handling multiple input types, supporting three languages, providing fast inference, enabling fine-grained aspect-level classification, and offering clear explanations. The paper ends with a detailed design plan for the proposed system, based on the findings from the literature review.

Keywords: Opinion Mining, Multimodal Sentiment Analysis, Explainable AI, Cross-Lingual NLP, Transformer Models, Social Media Analytics, Real-Time Classification

Introduction

Social media platforms have dramatically changed how quickly and widely public opinions form and spread. People from all around the world continuously post their reactions, reviews, and

comments, creating data streams that contain rich emotional information about products, events, policies, and personal experiences. Automatically identifying the sentiment of this content—whether it's positive, negative, or neutral—has become a highly important task both in terms of practical applications and scientific research. This field, known as sentiment analysis or opinion mining, is part of Natural Language Processing and has been actively studied since the early 2000s (Liu and Zhang, 2012). Commercial uses include brand monitoring, gathering customer feedback, predicting stock market trends, and tracking public health. However, a careful look at the current tools in sentiment analysis shows that they still have major shortcomings that limit how useful they are in real-world situations. Most widely used systems take only text as input, even though social media posts often include photos, short videos, and audio. By limiting analysis to text, these systems miss out on a lot of the emotional signal that's available. Another issue is that the focus is almost always on English. Countries like India, where the online population is one of the largest and fastest-growing globally, mostly produce content in Hindi, Marathi, and other regional languages, often mixing them in a single post. Most of the commercially available sentiment tools have little or no support for these linguistic realities. A third and more critical problem is the opacity of deep learning sentiment models. While these models often score well on benchmarks, they function like black boxes, assigning labels without explaining their reasoning, making it hard for users to understand, check, or trust the results. X-Sense is designed to overcome all three of these limitations within a single platform. The system will accept text, image, and audio-video inputs, support English, Hindi, and Marathi—including code-mixed content—and perform classification in real time. It will also include an integrated Explainable AI layer that provides human-readable explanations for every prediction. This paper is the first stage of the X-Sense development process. Its goal is to review the relevant literature systematically, compare existing approaches with the requirements of X-Sense, and create a justified design plan for the proposed system. The paper is structured as follows: Section 2 outlines the motivation; Section 3 presents the problem statement; Section 4 provides a literature review; Section 5 describes the proposed methodology; and Section 6 concludes with observations on future work.

Motivation

The motivation behind X-Sense comes from the noticeable gaps between what social media sentiment analysis needs and what current tools can offer.

Four particular gaps stand out.

Multimodal Communication as the Default Mode: Expressing sentiment online doesn't rely solely on written words anymore. Users often add photos, record audio, or upload videos to their posts. These different forms of media carry emotional information that text-only models fail to capture. A

comprehensive sentiment system needs to consider all available channels and use their combined signals, rather than focusing on just one.

Linguistic Diversity of Indian social media: India is one of the countries with the fastest-growing online populations, yet the infrastructure for analyzing sentiment in Indian social media is far behind the volume of content being produced. Hindi and Marathi, along with many other regional languages, are used daily by hundreds of millions of people who are largely invisible to English-only sentiment tools. The additional challenge of code-mixing, where users switch between Hindi, Marathi, and English within a single sentence, adds further complexity for any system aiming for broad applicability in Indian social media.

The Accountability Deficit in Black-Box AI: When organizations like news agencies, government health departments, or financial institutions use automated sentiment scores to make decisions, the credibility of these decisions depends on the ability to explain how the scores were generated. Deep learning models that only provide a label without explanation create an accountability gap. This isn't just an academic concern—it has real practical and sometimes legal consequences. Ensuring explainability from the start is therefore an essential design requirement in X-Sense, not an afterthought.

Absence of a Unified Solution: A detailed look at what has been published shows that no current system combines the ability to process different types of input, supports both Hindi and Marathi, provides real-time results, and explains how predictions are made, all in one platform. This is the main reason X-Sense was created.

Problem Statement

X-Sense tackles a challenge that involves four specific issues that existing systems haven't solved together.

Single-Modality Processing: Most sentiment analysis tools focus only on text. However, social media posts often include images, tone of voice, or video context that carry the main emotional message, not just the text. A system that ignores this non-text elements may misclassify posts where the visual or auditory signals contradict or add to what the text alone says.

Inadequate Multilingual Coverage: Most sentiment models are trained and tested on English data, and perform worse on other languages. Hindi and Marathi get little attention in research. Also, social media posts often mix languages, which monolingual systems can't handle.

Opaque Model Decisions: Top deep learning models make predictions without explaining the features or reasoning behind them. People using these models for important decisions can't check if

the model is responding to real sentiment or just coincidental patterns in the training data. Without explanations, trust and accountability suffer.

No Integrated Real-Time Platform: There is no single tool that handles multimodal analysis, supports multiple languages, delivers real-time results, and offers explanations. X-Sense is specifically designed to provide all these features in one system.

Literature Survey

4.1 Lexicon-Based Approaches

Early work in sentiment analysis used pre-built lists of words that indicated positive or negative feelings. These lists, like SentiWordNet, often start with a small set of manually selected words, then expand by looking for synonyms and antonyms in databases like WordNet. Other methods, called corpus-based, use large amounts of text to find patterns of how words appear with known sentiment words. A comparison of available lexicon tools found that the Valence Aware Dictionary and Sentiment Reasoner performs better, especially on informal content like social media, due to handling features like uppercase letters and punctuation. However, all lexicon-based methods struggle with understanding how the same word can mean different things in different contexts. Language mixing, regional slang, and new internet terms make this even harder, limiting these tools to well-structured, stable content.

4.2 Classical Supervised Machine Learning

Supervised machine learning treats sentiment classification as a standard text categorization task. Documents are converted into numerical data, often using methods like Bag of Words, TF-IDF, or n-gram frequencies, and a classifier is trained to assign sentiment labels using labeled examples. The Support Vector Machine, developed by Cortes and Vapnik, is a popular method that builds clear boundaries between different classes. It is widely used in sentiment analysis because it works well with high-dimensional text data. Saravanan et al. reviewed various classical algorithms for sentiment tasks. Naive Bayes is praised for being fast and effective with limited data. Logistic Regression provides clear probability estimates but is limited to linear decisions. Decision Trees are easy to understand and build but tend to overfit. K-Nearest Neighbours classifies data by looking at similar examples but slows down with large datasets. All these methods rely on handcrafted features that capture word frequency but not meaning, grammar, or long-term context.

Table 1: Comparison of Conventional Sentiment Analysis Methods

Method	Primary Strength	Primary Weakness
Lexicon (VADER)	Requires no labelled training data	Cannot resolve context-dependent polarity
Support Vector Machine	Strong on high-dimensional sparse features	Training time scales poorly with data volume
Naive Bayes	Efficient with small training sets	Conditional independence assumption rarely holds
Logistic Regression	Outputs calibrated probability estimates	Decision boundaries limited to linear separations
Decision Tree	Human-readable classification logic	High susceptibility to overfitting
K-Nearest Neighbours	No explicit training phase needed	Slow inference on large feature spaces

4.3 Deep Learning Architectures

Using deep neural networks for sentiment analysis has made it possible to learn complex, layered text representations directly from data, without needing to manually create features.

Chen (2015) showed that a Convolutional Neural Network using word2vec embeddings could match or even beat the performance of classifiers that rely on manually created features on sentence-level sentiment tasks. Convolutional models are great at finding local patterns in text, like common word groups, but they are not as good at understanding relationships between words that are far apart in a sentence. Recurrent Neural Networks were developed to handle sequential data by keeping track of information from earlier parts of the input. However, standard RNNs struggle with long sequences because the training process can cause important information to be lost or become too strong (Zaremba et al., 2014). The Long Short-Term Memory (LSTM) architecture, introduced by Hochreiter and Schmidhuber (1997), solves this by using a gate system that allows the network to control what information to keep, update, or forget as it processes each word. Bidirectional LSTM takes this a step further by processing the text in both directions at the same time, letting each word have access to both the words before and after it. Mao et al. (2022) used an attention-augmented Bidirectional LSTM with a 2D convolutional network for document-level classification and found that combining these models can lead to better results than using either alone. The Gated Recurrent Unit (GRU) offers a faster alternative to LSTM by combining some of the gate

operations, which means it uses less computation but might not capture as much information as LSTM, especially on smaller datasets.

4.4 Transformer-Based Pre-Trained Models

The Transformer architecture, which relies entirely on attention mechanisms instead of recurrent structures, made it possible to create new types of language models that can be trained on huge amounts of unlabeled text and then adapted to specific tasks with just a small amount of labeled data. Devlin et al. (2018) introduced BERT, which stacks multiple layers of bidirectional Transformer encoders and is trained using two main tasks: predicting masked words and determining if two sentences follow each other. Fine-tuning BERT on sentiment analysis tasks leads to top results across standard benchmarks, and its bidirectional nature makes it more aware of context than models that process text in one direction only. Several versions of BERT have been developed to fix specific issues with the original. Liu et al. (2019) created RoBERTa, which performs better by training with larger batches, more data, and longer sequences, while removing the task of predicting whether sentences follow each other. Clark et al. (2020) introduced ELECTRA, which changed the masked language modeling task to a token discrimination task, achieving similar accuracy with less computational cost. Liu et al. (2020) added structured knowledge from knowledge graphs to BERT in their K-BERT model, improving performance on tasks that benefit from domain knowledge. For multilingual use, mBERT extends the model to over 100 languages by training it on multiple languages at once, and XLM-RoBERTa builds on this by applying RoBERTa's improvements in a multilingual context. A common challenge with all BERT-based models is their large number of parameters, with the base version having around 110 million weights and requiring significant computational power. For real-time applications like X-Sense, techniques like knowledge distillation, such as DistilBERT, reduce the model size and speed up inference while keeping most of the performance.

4.5 Multimodal Sentiment Analysis

Gandhi et al. (2023) did a detailed review of multimodal sentiment analysis, listing datasets, fusion methods, and open research areas. Their survey showed that systems that use multiple types of input—text, audio, and video—perform better on the CMU-MOSI and MOUD benchmarks than systems that rely on just one type of input. Three main fusion strategies have been explored in research: early fusion combines raw or low-level features from different modalities before feeding them into a single classifier; late fusion trains separate classifiers for each modality and then combines their predictions; and hybrid fusion, which merges intermediate representations instead of raw features or final predictions, tends to be the most effective at capturing interactions between

different modalities. Attention-based and tensor-based hybrid approaches are especially good at adjusting the weight of each modality based on the input content. One major limitation noted by Gandhi et al. is that almost all multimodal sentiment research has focused on English, making cross-lingual multimodal analysis a largely unexplored area that is highly relevant to X-Sense.

4.6 Multilingual and Code-Mixed Text

Saadany and Orasan (2020) studied the challenges of keeping sentiment accurate during cross-lingual machine translation, finding that translation systems often distort sentiment because they handle negations, intensifiers, and idiomatic expressions poorly. This raises doubts about the reliability of approaches that translate non-English text into English before applying a single-language sentiment model. Mao et al. (2024) reported that while pre-trained multilingual models like XLM-RoBERTa have good zero-shot transfer capabilities across languages, their performance on low-resource and code-mixed data is much worse than what is possible with standard English inputs. The challenge of analyzing sentiment in Hindi-English-Marathi code-mixed text has received very little focused research, and the available tools for this specific context are limited.

4.7 Explainable AI for Sentiment Systems

Arrieta and their team in 2020 created a detailed list of Explainable AI methods, sorting them based on how interpretable they are, which type of model they work with, and who they are meant to explain things to. Ribeiro and their group in 2016 came up with LIME, which gives a clear explanation for a single prediction by changing the input slightly, seeing how the model's output changes, and then building a simpler, more understandable model that matches this local behavior. When using text, LIME gives a list of words ranked by how much they influenced the prediction. The SHAP framework, which is based on Shapley values from game theory, offers another method that gives each input feature a score showing its contribution, ensuring fairness and completeness so that both local and overall model behavior can be explained. GradCAM uses gradient data from the model to create heatmaps that show which parts of an image were most important for a classification, making it useful for explaining image-based decisions. A review of published sentiment analysis systems shows that explainability features are not often included in real-world tools, and no system currently covers explainability across text, image, and audio all at once.

Methodology

The X-Sense system is designed as a modular, layered system where each part works with the ones next to it through clear, defined interfaces. This lets each part be developed, tested, replaced, or updated without messing up the whole system. The system has five main layers: data input and

preparation, multilingual processing, multimodal feature extraction, sentiment combination and classification, and explainability and visualization.

5.1 Data Ingestion and Pre-processing

Content enters the system through four ways: typed or pasted text, uploaded images, uploaded audio or video files, and social media URLs from Twitter or Instagram. Audio and video files go through a preprocessing step where the audio is separated and video frames are taken at regular intervals for visual processing. Text is cleaned by removing HTML tags, standardizing Unicode characters including Devanagari script, converting to lowercase, and removing special characters that don't affect sentiment. A language detection tool using the langdetect library gives each text input a language tag. If the language is Hindi or Marathi, the text is sent to a translation component using Google Translate or the IndicNLP library to create an English version for the model, while keeping the original text to support accurate explanations.

5.2 Multimodal Feature Extraction

Three specialized encoders are used to create features for each modality. For text, a fine-tuned XLM-RoBERTa model makes a 768-dimensional contextual embedding for each input. XLM-RoBERTa is used instead of standard multilingual BERT because it includes improvements from RoBERTa trained on a multilingual corpus with 100 languages, which leads to better performance on low-resource languages like Hindi and Marathi. The model is fine-tuned using a dataset combining SemEval benchmarks, public Hindi Twitter sentiment sets, and a Marathi social media collection made for this project. Visual sentiment features are taken using a ResNet-50 network trained on ImageNet. The original classification layer is replaced with a fully connected layer trained on the Getty Images Sentiment Dataset and the Twitter Image Sentiment collection, aligning the visual features with emotional categories. For audio content, the librosa library extracts features like Mel-frequency cepstral coefficients, chroma, and spectral contrast from the audio, which are then given to a lightweight CNN-based audio sentiment classifier. Video content is handled by feeding sampled frames through the image encoder and the audio track through the audio encoder, then combining the feature vectors during the fusion stage.

5.3 Multimodal Fusion and Sentiment Classification

The three sets of features from each modality are combined using a cross-modal attention mechanism where each modality pays attention to the relevant parts of the other two. This mixed method has been found to perform better than simply combining features in the multimodal sentiment research (Gandhi et al., 2023). The resulting fused data goes into a three-class sentiment classifier that is a multi-layer perceptron with dropout to avoid overfitting, giving probabilities for

positive, negative, and neutral sentiments. Training uses cross-entropy loss with class weighting to handle the class imbalance common in social media datasets. The classifier is built using PyTorch and served through a FastAPI backend set up for asynchronous inference to support real-time performance.

5.4 Explainable AI Layer

Each prediction made by X-Sense comes with a structured explanation displayed on the results dashboard. For text inputs, LIME creates token-level scores by changing the input and checking how the prediction probability changes. The top five tokens that most affected the classification are shown in color: green for positive contributions and red for negative ones. SHAP values are calculated at the same time to give a global ranking that puts the local explanation into context. For images, GradCAM creates a heatmap showing the image areas most responsible for the sentiment. For audio, temporal SHAP values identify the parts of the sound that influenced the classification. The explanation part is added as a step after prediction, making sure it completes within a 500 millisecond delay.

5.5 Technology Stack

Table 2: Proposed Technology Stack for X-Sense

Component	Technology / Library	Purpose
Programming Language	Python 3.10	Primary development environment
Deep Learning Framework	PyTorch 2.0	Model construction and training
Text Encoder	XLM-RoBERTa (HuggingFace)	Multilingual contextual embeddings
Image Encoder	ResNet-50 (torchvision)	Visual sentiment feature extraction
Audio Processing	librosa, wav2vec2	Acoustic feature computation
Explainability	LIME, SHAP, GradCAM	Prediction attribution and explanation
Backend API	FastAPI + Uvicorn	Asynchronous real-time model serving
Database	MongoDB	Storage of inputs, results, and users

Component	Technology / Library	Purpose
Frontend	HTML5, CSS3, JavaScript, Chart.js	Interactive results dashboard
Social Media APIs	Twitter API v2, Instagram Graph API	Live content ingestion
Containerisation	Docker	Portable cross-environment deployment

5.6 End-to-End Workflow

A registered user logs in through the web app and can submit content using one of four ways: typing text directly, uploading an image, uploading audio or video, or sharing a link from a social media platform. The system checks what language the content is in and translates any non-English text before it goes through the sentiment analysis model. Special tools called modality-specific encoders create detailed descriptions of the input, which are then combined and analyzed by the multimodal model. A separate module called XAI creates explanations that show why the model made its prediction. The dashboard shows the sentiment result, how confident the model is, highlights important parts of the input that influenced the prediction, and provides visual summaries like a pie chart showing the likelihood of different sentiment types and a bar chart showing which parts of the text or image were most important. Users can download a PDF report that includes all the input content, the model's predictions, the explanations, and the visual charts. All the results are stored in a MongoDB database for future access and comparison over time.

Conclusion

This first stage of the review has looked closely at the current state of sentiment analysis research and mapped out how the findings fit with the design needs of X-Sense. The survey shows that methods like using word lists, traditional supervised learning, deep learning, transformer models, and multimodal approaches each bring improvements in one or more areas. However, none of these approaches, either on their own or together, meet all five requirements at the same time. The lack of labelled sentiment data in Marathi and in code-mixed Hindi-English-Marathi is a major challenge, and it will be addressed using back-translation and cross-lingual learning techniques. Detecting sarcasm and subtle emotions remains difficult, but context-aware attention mechanisms and models like K-BERT can help to some extent. To handle the computational issues of real-time multimodal analysis, techniques like model distillation and asynchronous API design will be used. For aspect-level models that perform worse in different domains, multi-domain fine-tuning and domain-

adaptive pre-training will be used to improve their performance. Stage Two will turn this architectural plan into a working system by gathering training data, fine-tuning the individual models, putting the entire workflow together, and testing the system against existing benchmarks. The thorough analysis and methodological setup from this review provide a strong reason for each design decision and set clear performance expectations for the next step.

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